

distribution of the inclusions correspond to those of lipofuscin; 3. The inclusions become more prominent with age or duration of the disease like ordinary lipofuscin granules; 4. The quantity of the inclusions does not exceed that of lipofuscin granules which are expected to occur in neurons of individuals of similar age with other long-standing illnesses; 5. The inclusions vary somewhat in size and shape from one nerve cell to another and from one nuclear group to another.

The salient difference between the neuronal inclusions in Chediak-Higashi disease and ordinary lipofuscin granules in other conditions is in their size and shape. Lipofuscin granules are small, discrete and fairly uniform, whereas the inclusions are large, irregular in shape, and are formed as a result, most likely, of agglomeration or fusion of lipofuscin granules. Similarly, the pigment granules of the nerve cells of the substantia nigra in patients with Chediak-Higashi disease differ from those in other individuals of similar age.

Thus, whatever it is that maintains the integrity, discreteness and individuality of lipofuscin granules and other pigment granules of nerve cells is apparently abnormal in Chediak-Higashi disease. This abnormality seems to exist in the single membrane bound subcellular organelles which give rise to lipofuscin granules. The subcellular organelles are believed to be lysosomes. This abnormality is apparently genetically induced. Conditions causing lipofuscin granules or secondary lysosomes to occur seem to make the abnormality manifest through the production of the inclusions.

PAS positive granules were frequently encountered in practically all types of cells including astrocytes, choroid plexus, epithelium, Schwann cells, fibrocytes, histiocytes, and other inflammatory cells.

Amyloid in Late Postradiation Necrosis of Brain. Tadeusz J. Mandybur, and Ira Gore (by invitation), *Boston, Mass.*

This communication concerns the neuropathological changes in 2 patients who received accidental irradiation of the brain when treated for lesions of the scalp.

Late postradiation necrosis of this brain is a well-defined entity, but some histological features, such as presence of "amyloid-like material" in the lesions, have been described in only a few cases. The pathogenesis of the appearance of "amyloid" in these cases is obscure. In cases from the literature, the material which stained with Congo red, failed to stain with Lugols solution, methyl violet, and/or thioflavin T. Because green birefringence had not been observed with polarized light, this was often not considered to be true amyloid.

Using a polarizing microscope, rather than simple filters, both of our cases displayed green birefringent material after Congo red staining. This amyloid appeared interstitially as amorphous masses, in and about blood vessels producing forms resembling dyschoric angiopathy.

The accumulations in Case I, at autopsy, were found to be limited to a portion of the irradiated field displaying ischemic necrosis. The role of ischemia in previously irradiated brain in the development of amyloid is considered.

An experimental approach to the problems of amyloid in postradiation necrosis of brain is discussed.

DISCUSSION

DR. W. ZEMAN, *Indianapolis, Indiana*: This was a rather nice demonstration of traumatic brain edema following focal X-irradiation. There are several points that I would like to raise. I do not believe that the focal dose in the brain was 16,000 R, since this was very soft radiation and the absorption coefficient of bone is in the range of 2/cm. The observations on the amyloid, or amyloid-like substance are interesting, but it is difficult to assess these findings, not so much because of the intricacies of staining and of fluorescent reaction, but also because we have no good identification or definition of amyloid. For this reason I would like to ask whether you stained fresh frozen sections with crystal-violet?

Did you do any sulfation tests? Was there any evidence that glycoproteins were really involved? Was this material PAS positive?

DR. T. MANDYBUR: Amyloid is detected by its staining characteristics and appears as hyaline substance with certain unique staining and optical reactions. Of all the methods used, the most important are those directed at demonstrating congophilia and green birefringency. All other staining reactions are of lesser importance; I am not personally acquainted with the sulfation reaction. Cresyl-violet for example proves to be negative in mice with amyloidosis induced experimentally. Yet this is clearly amyloid as shown by electron microscopy. It, nonetheless, never reveals metachromasia with the cresyl-violet stain.

The Infantile Spongy Degenerations. Irwin Feigin, Gleb Budzilovich, and Carlos Peña (by invitation), *New York, N.Y.*

One form of spongy state is thought to result from the agonal or postmortem evolution of carbon dioxide gas into the tissues, derived from glycogen or other carbohydrate reservoirs by the persistence for a short period, of the normal or of altered decarboxylating mechanisms. The brains of 7 infants with considerable spongy changes were reviewed, and the sponginess in these was judged to be of this character. In 3, the metabolic abnormalities of maple syrup urine disease were demonstrated. One contained lesions of subacute necrotic encephalomyelopathy, in addition to areas of spongy change. The other 3 were grouped with Canavan-van Bogaert group. The spongy changes were present in areas of myelin deficiency, but also in normally myelinated tissues, particularly in the brain stem. In some of the cases, present or antecedent edema was evident histologically in some of the spongy areas, and also in some non-spongy areas. Edema was not demonstrated in other cases, and was demonstrably absent by histologic criteria in some markedly spongy tissues.

It is considered that the sponginess *per se* does not represent edema and does not determine an entity, but may occur in various infantile and other diseases with metabolic abnormalities.

DISCUSSION

DR. J. AUSTIN, *Denver, Colorado*: Would you tell us if there is any correlation between the appearance of this sponginess and the interval between death and autopsy, or the length of time before the body was refrigerated? As I understand it, your belief is that this is carbon dioxide which may be the result of glycogen breakdown. One wonders if this glycolytic activity is enhanced at a higher temperature.

DR. M. IBRAHIM, *Moffett Field, California*: I would like to mention 2 points. First regarding astrocytes in the U-fiber layer: In my experience, the astrocytes in the U-fiber layer are always much better and more consistently impregnated with the Cajal method than the astrocytes in either the gray or the white matter. Recently, we completed a paper on 2 artefacts of fixation, which we found upon fixing brain in alcohol-formalin and formalin. These artefacts appeared as small metachromatic bodies. These bodies we found histochemically were composed of sulfatides and, perhaps, also acid mucopolysaccharides. When these bodies dissolve out in the preparation of paraffin section this leaves holes such as seen here. Of course, they are not quite as numerous as has been shown here tonight.

DR. IRWIN FEIGIN: I regret, Doctor Austin, that I can not give you that specific information with respect to these 7 children. We studied this point in our original group of cases and I can indicate that it does not make very much difference how long the patient had been dead prior to autopsy, at least up to 36 hours. Some glycogen persists for this period. The question concerning refrigeration, I can not answer.

Doctor Ibrahim, with respect to the morphology of astrocytes in normal tissues, our experience seems to differ from yours. We believe that the astrocytes of the arcuate zone normally differ in no respect, not in number, distribution, or morphology, from those more deeply placed. We feel that they do differ consistently in edema. The other mechanisms which you suggest, I feel do not apply to this situation, although they may apply to other